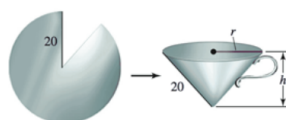


Problem 6

Problem
metal
and h



Explanation. First, recall that the volume of a cone is

$$V = \frac{1}{3}\pi r^2 h \quad (1)$$

Equation (1) has two variables, r and h . So we need to find a constraint equation. Notice that the length of the cone is a fixed length of 20. So using Pythagorean's Theorem we have that:

$$r^2 + h^2 = 20^2 = 400 \quad (2)$$

Solving equation (2) for r^2 , we get $r^2 = 400 - h^2$. Plugging this into equation (1) gives

$$V = \frac{1}{3}\pi(400 - h^2)h = \frac{1}{3}\pi(400h - h^3) \quad (3)$$

Note that, due to the settings of the problem, the domain for h is $[0, 20]$.

It is worth pointing out that it is a lot easier algebraically to solve for r^2 in equation (2) and then plug into equation (1) instead of doing the same thing for h .

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Now we need to differentiate equation (3) with respect to h , set this equal to 0, and solve for h .

$$\frac{dV}{dh} = \frac{1}{3}\pi(400 - 3h^2) := 0$$

$$400 - 3h^2 = 0$$

$$3h^2 = 400$$

$$h^2 = \frac{400}{3}$$

$$h = \pm\sqrt{\frac{400}{3}} = \pm\frac{20}{\sqrt{3}}$$

but $-\frac{20}{\sqrt{3}}$ is not in our domain for h , and so the only critical point for $V(h)$ is $h = \frac{20}{\sqrt{3}}$. We need to show that this is a global maximum for $V(h)$ on $[0, 20]$. Since $[0, 20]$ is a closed interval, the Extreme Value Theorem says we need to evaluate $V(h)$ at $h = 0, \frac{20}{\sqrt{3}}, 20$.

$$V(0) = \frac{1}{3}\pi(0 - 0) = 0 \tag{4}$$

$$V\left(\frac{20}{\sqrt{3}}\right) = \frac{1}{3}\pi\left(\frac{20^3}{\sqrt{3}} - \frac{20^3}{(\sqrt{3})^3}\right) > 0 \tag{5}$$

$$V(20) = \frac{1}{3}\pi(20^3 - 20^3) = 0 \tag{6}$$

Thus $h = \frac{20}{\sqrt{3}}$ maximizes the volume of the cone. Then

$$r^2 = 400 - h^2 = 400 - \left(\frac{20}{\sqrt{3}}\right)^2 = 400 - \frac{400}{3} = \frac{800}{3}$$

and so

$$r = 20\sqrt{\frac{2}{3}}.$$

If you are not comfortable with inequality (5) above without a calculator, then a nice alternative is to use the second derivative test instead to check that $h = \frac{20}{\sqrt{3}}$ maximizes the volume of the cone. This works since this is the only critical point in the domain of h . To do this, compute

$$\frac{d^2V}{dh^2} = \frac{1}{3}\pi(-6h)$$

$$\left[\frac{d^2V}{dh^2}\right]_{h=\frac{20}{\sqrt{3}}} = \frac{1}{3}\pi\left(-6\left(\frac{20}{\sqrt{3}}\right)\right) < 0$$

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and thus this value for h gives a local (and therefore, global) maximum value for $V(h)$.